AI Glasses Carrier V1 - Power Budget & Battery Sizing

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Date	**2026-06-30**

- **Generated:** 2026-06-30 | Source: `config/power_budget.yaml`
- **Assumptions:** 1S Li-Po 3.7 V nominal; boost eff 90%; usable DoD 90%.
- **[!] ESTIMATE** from datasheet typicals - NOT bench-measured. Fill measured current (R3 0R/shunt on +5V) during bring-up and re-run.

1. Per-subsystem load power (mW)

Subsystem	Idle	Typical	Full	Note
CM4 (RK3576 SoC + LPDDR4X + eMMC)	450	2000	2900	Dominant load; NPU AI + camera ISP + H.265 encode drive the peaks
Wi-Fi / BT radio (on CM4)	50	200	750	TX bursts; radio is on the CM4 module
Camera IMX415 (4-lane MIPI)	0	400	450	Off in idle; streaming otherwise
Audio amp + speaker (avg)	0	150	300	Class-D average for speech; short music peaks to ~3W not in the average
PDM mics x2	10	20	20	
GNSS MAX-M10S	20	80	80	~25 mA @3.3V during acquisition/tracking
IMU ICM-42688-P	1	1	2	Low-power 6-axis
Haptic DRV2605L (avg)	0	0	100	Only during a vibration pulse
Status LEDs	20	30	30	
LDO quiescent + regulation loss margin	100	150	200	3V3/1V8 LDO drop + headroom
****TOTAL load****	**651**	**3031**	**4832**	mW

2. Battery-side current & runtime

Battery current = load / (3.7 V x 90% boost). Runtime uses 90% usable DoD.

Scenario	Load (W)	Batt current (mA)	Draw/hour (mAh)	V1 bench 103450 (2000 mAh)	Glasses pack (2-3 cells) (650 mAh)
Idle	0.65	195	195	9.2 h	3.0 h
Typical	3.03	910	910	2.0 h	0.6 h

Scenario	Load (W)	Batt current (mA)	Draw/hour (mAh)	V1 bench 103450 (2000 mAh)	Glasses pack (2-3 cells) (650 mAh)
Full	4.83	1451	1451	1.2 h	0.4 h

3. Battery capacity required for a target runtime

Capacity (mAh) needed to sustain each scenario for the target time (1S 3.7 V, 90% DoD, boost 90%):

Scenario	Load (W)	1 h	2 h	3 h	4 h
Idle	0.65	217 mAh	434 mAh	652 mAh	869 mAh
Typical	3.03	1011 mAh	2023 mAh	3034 mAh	4045 mAh
Full	4.83	1612 mAh	3225 mAh	4837 mAh	6449 mAh

4. Bottom line

- **Typical use ~3.0 W** -> battery draws **~910 mA**, i.e. **~1011 mAh per hour**.
- Hitting the **3-4 h** target at typical load needs **~3034-4045 mAh**.
- **V1 bench 103450 (2000 mAh)** -> ~2.0 h typical, ~1.2 h full, ~9.2 h idle.
- **Glasses pack (2-3 cells) (650 mAh)** -> ~0.6 h typical, ~0.4 h full, ~3.0 h idle.
- The glasses pack cannot reach 3-4 h at a steady ~3 W; that target needs **AI duty-cycling** (wake-on-demand, gate the NPU/camera/encode), not just a bigger cell.

Estimates from `config/power_budget.yaml`. Replace with R3-shunt bench measurements during bring-up and re-run `scripts/power_budget.py`.